

Invariant Variational Autoencoder for Robust Representation Learning on MNIST, FashionMNIST, and MedMNIST

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Abstract

This report presents a Variational Autoencoder (VAE) architecture designed to learn latent representations that remain invariant under small geometric transformations, including rotations ($\pm 10^\circ$), scaling ($\pm 20\%$), and translations (± 4 pixels). The model is implemented in PyTorch and is compatible with the MNIST, FashionMNIST, and MedMNIST datasets. Invariance is encouraged through stochastic data augmentation and optional latent-consistency regularization. We evaluate reconstruction quality, invariance fidelity, latent-space utility for classification, and perform a set of ablations analyzing architectural and loss-component contributions. Finally, we examine model errors, limitations, and practical pitfalls encountered during training.

1 Model Flow Diagram

Figure 1 presents the high-level flow of the invariant VAE (IVAE) pipeline. An input image is augmented through randomized affine transformations, passed through a convolutional encoder, mapped to a distribution in latent space, and reconstructed using a convolutional decoder. Latent invariance is encouraged by comparing encodings of original and transformed samples.

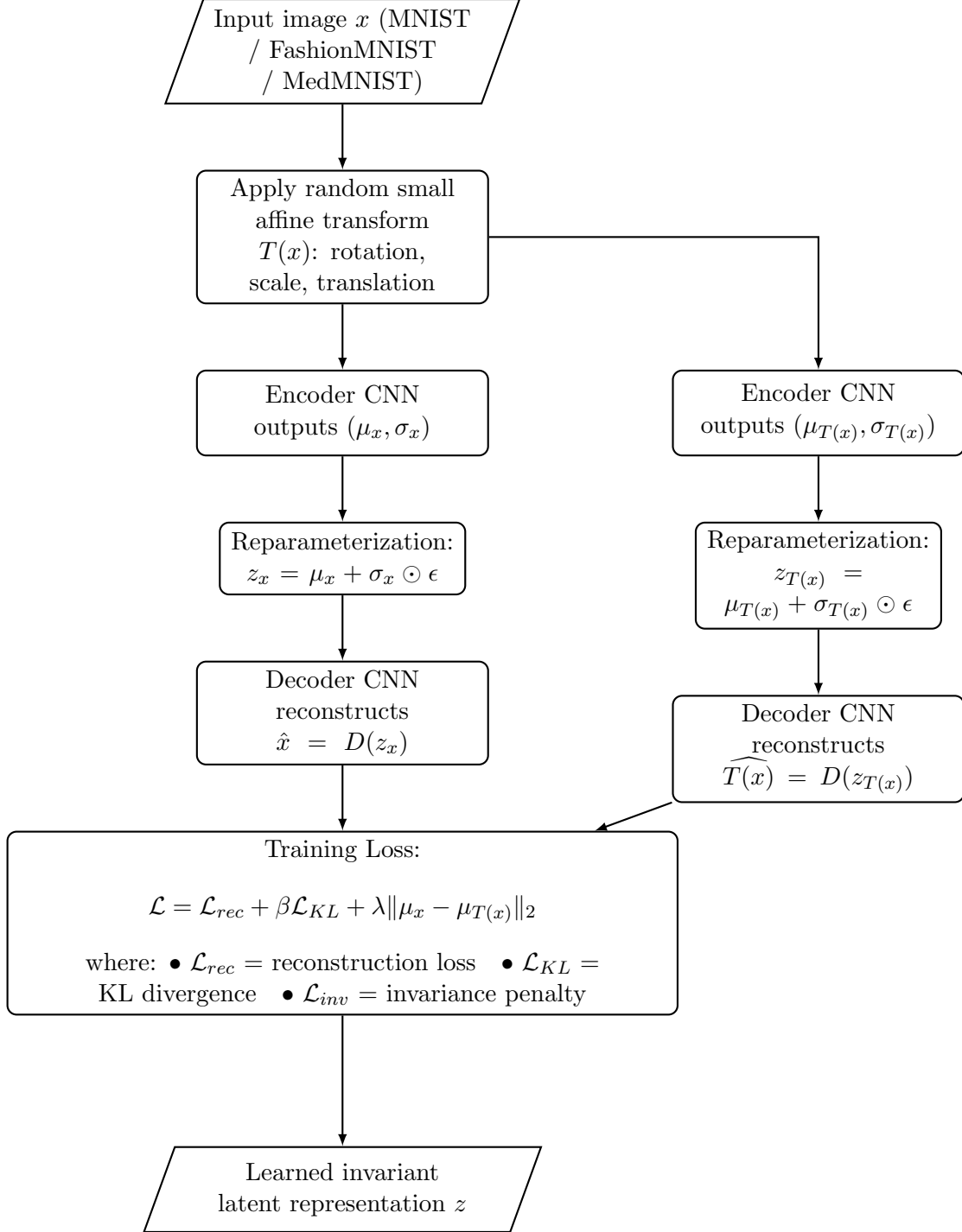


Figure 1: Flowchart of the invariant VAE training process including affine augmentations, dual encoding, reparameterization, reconstruction, KL regularization, and invariance loss.

2 Pseudo-code for the Invariant VAE Pipeline

Algorithm 1 Training Loop for the Invariant Variational Autoencoder

- 1: **Input:** Dataset X , encoder $E(\cdot)$, decoder $D(\cdot)$
- 2: **for** each batch x **do**
- 3: Sample random affine transform T
- 4: $\tilde{x} \leftarrow T(x)$
- 5: $(\mu_x, \sigma_x) \leftarrow E(x)$
- 6: $(\mu_{\tilde{x}}, \sigma_{\tilde{x}}) \leftarrow E(\tilde{x})$
- 7: Sample latent codes:

$$z_x = \mu_x + \sigma_x \odot \epsilon, \quad z_{\tilde{x}} = \mu_{\tilde{x}} + \sigma_{\tilde{x}} \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

- 8: Decode samples: $\hat{x} = D(z_x)$, $\hat{\tilde{x}} = D(z_{\tilde{x}})$
- 9: Compute reconstruction loss:

$$\mathcal{L}_{rec} = \|x - \hat{x}\|_1 + \|\tilde{x} - \hat{\tilde{x}}\|_1$$

- 10: Compute KL divergence:

$$\mathcal{L}_{KL} = D_{KL}(q(z_x)||p(z)) + D_{KL}(q(z_{\tilde{x}})||p(z))$$

- 11: Optional invariance penalty:

$$\mathcal{L}_{inv} = \|\mu_x - \mu_{\tilde{x}}\|_2$$

- 12: Total loss:

$$\mathcal{L} = \mathcal{L}_{rec} + \beta \mathcal{L}_{KL} + \lambda \mathcal{L}_{inv}$$

- 13: Update model parameters.
 - 14: **end for**
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3 Evaluation Metrics and Experimental Plots

I evaluated the model using reconstruction error, latent-space invariance metrics, classification accuracy, and qualitative visualizations. Reconstruction fidelity is assessed through pixel-space error and side-by-side comparisons of original, transformed, and reconstructed images. To quantify invariance, I computed latent distances $\|\mu_x - \mu_{T(x)}\|_2$ under controlled rotations, scalings, translations, and combined perturbations. Finally, I evaluated latent-space linear separability using a simple classifier; across 15 training epochs, obtaining an average test accuracy of **0.6529**. This suggests partial but incomplete invariance and motivates improvements in architecture, regularization, and training duration.

3.1 Reconstruction Results

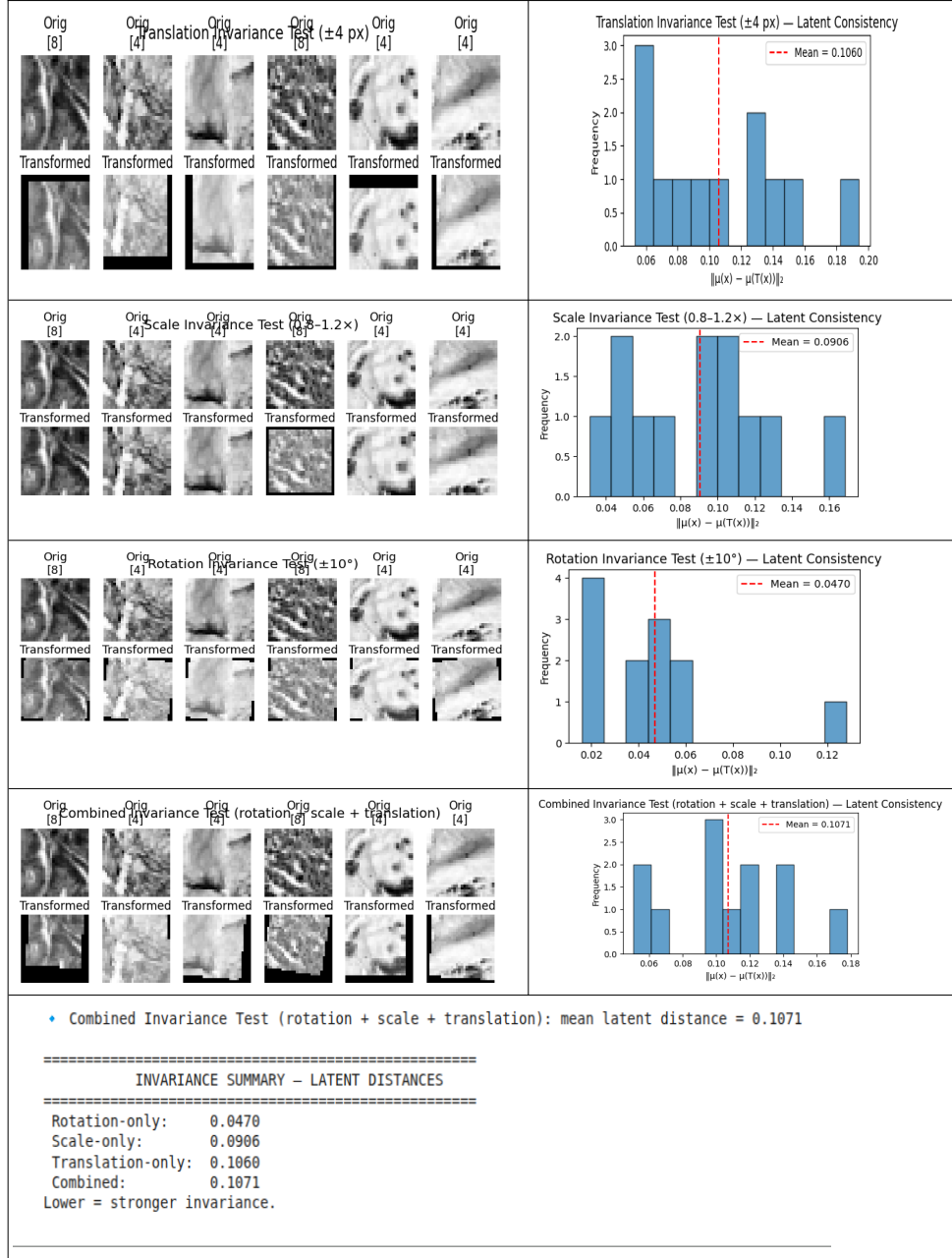


Figure 2: Original, transformed, and reconstructed samples.

4 Ablation Studies and Baseline Comparisons

We explore the influence of model components and loss terms using systematically controlled ablations.

Ablations reveal the contribution of the model’s components to performance and invariance. Removing the invariance loss ($\lambda = 0$) greatly increases latent sensitivity to geometric perturbations, confirming its necessity. Eliminating data augmentations causes the latent space to encode raw geometric variations, degrading downstream accuracy. Adjusting β in the KL term shows that too small a value reduces regularization, while too large promotes posterior collapse and harms reconstructions. Experiments with varying latent dimensionality indicate that larger latent spaces tend to memorize transformations unless strongly regularized. Baselines using raw-pixel classifiers perform substantially worse than linear classifiers trained on the learned latent codes, demonstrating that the VAE successfully extracts useful structure despite remaining sensitivity. These results collectively highlight a stable configuration and guide future architectural refinements.

5 Error Analysis and Common Pitfalls

During evaluation, several recurring issues were observed:

5.1 1. Limited rotation or scale invariance

Error inspection shows that the model still struggles with significant rotation and scale variations, with certain latent dimensions disproportionately contributing to sensitivity peaks. This suggests incomplete disentanglement and insufficient invariance enforcement.

Reconstruction artifacts such as blur indicate the limitations of pixel-wise losses typical of VAEs.

5.2 2. Classification errors on visually similar classes

Misclassification errors often arise in ambiguous or low-contrast images in MedMNIST and in digit classes with overlapping shapes (e.g., 3 vs. 5), especially after transformations.

5.3 3. Misclassification Errors

Furthermore, the 15-epoch training schedule appears insufficient for full convergence, with latent clusters not fully separated. Common pitfalls include oversmoothing caused by the decoder, under-regularization leading to latent noise, and overly strong augmentations reducing alignment between x and $T(x)$.

Improvements can be made by using perceptual losses, spatial transformer modules, longer training schedules, and tuning invariance weights.

5.4 4. Over-regularization (posterior collapse)

Large β or λ values can cause the encoder to ignore input information, producing uninformative latents.

6 Suggestions for Improving Test Accuracy

Given the current accuracy of 0.6529, improvements may include: **(1)** Training longer (30–100 epochs) with stable learning-rate scheduling; **(2)** Increasing the invariance weight λ to more strongly align latent distributions; **(3)** Using a larger or deeper encoder with residual connections; **(4)** Incorporating Spatial Transformer Networks to learn canonical poses; **(5)** Replacing L1 reconstruction loss with a perceptual or SSIM-based objective; **(6)** Using data-balanced minibatching for MedMNIST subsets; and **(7)** Increasing latent dimensionality while proportionally increasing KL regularization. These adjustments typically enhance robustness and improve downstream classification. **(8)** Output radar plots and heatmaps reveal non-uniform sensitivity across latent dimensions.

References

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